

Critical Artificial Intelligence literacy: A scoping review and framework synthesis

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ABSTRACT

The proliferation of Artificial Intelligence (AI) in everyday life raises concerns for children, other marginalized groups, and the general public. As new AI implementations continue to emerge, it is crucial to enable children to engage critically with AI. Critical literacy objectives and practices can encourage children to question, critique, and transform the social, political, cultural, and ethical implications of AI. As an initial step towards critical AI education, we conducted a 10-year scoping review to identify publications reporting on activities that engage children, between the ages of 5 and 18, to address the critical implications of AI. Our review identifies a wide range of participants, content, and pedagogical approaches. Through framework synthesis guided by an established critical literacy model, we examine the critical literacy learning objectives embedded in the reported activities and propose a critical AI literacy framework. This paper outlines future opportunities for critical AI literacies in the field of child–computer interaction including inspiring new learning activities, encouraging inclusive perspectives, and supporting pragmatic curriculum integration.

1. Introduction

Artificial Intelligence (AI) is becoming an increasingly entangled aspect of our everyday lives. With the gradual increase in computing power and the availability of large datasets, AI applications have seen a large uptake in a variety of fields. Young people are growing up exposed to AI in their daily lives not only through AI-driven products like toys, smartphones, and smart home devices, but also through experiencing societal impacts such as job automation (Frey & Osborne, 2017), privacy concerns (boyd & Crawford, 2012), and algorithmic bias perpetuating discrimination (Eubanks, 2018). Consequently, there is an urgency to conduct research on how to support children and youth in developing critical literacy for engaging with these new AI-powered technologies.

Within Child–Computer Interaction (CCI), the paradigm of Computational Empowerment focuses on empowering children to make critical and informed decisions about the role of technology in their lives (Iversen, Smith, & Dindler, 2018). From this perspective, education on emerging technology should not only impart technical skills to students, but also provide them with the ability to think critically about how technologies may constitute and/or contribute to social, political and ethical issues. Public policy and the research community both stress the importance of integrating ethical and sociopolitical considerations as part of AI literacy (Adams, Pente, Lernermeier, & Rockwell,

2023; Touretzky, Gardner-McCune, Martin, & Seehorn, 2019). For instance, Long and Magerko's (2020) AI literacy framework contains objectives around educating students about privacy issues, employment, diversity in the computer science workforce, algorithmic bias or accountability of AI. While there have been efforts in creating AI pedagogy to equip students with skills to thrive in an AI-driven world, systemic reviews of formal AI curricula reveal few learning activities emphasizing AI ethics or supporting youth reflection on personal and sociopolitical impacts of AI (Casal-Otero et al., 2023; Martin, Zhuang, & Schaefer, 2024; Rizvi, Waite, & Sentance, 2023).

We can support youths' Computational Empowerment regarding emerging AI technologies through *critical literacy (CL)* objectives and practices (Velandar, Otero, & Milrad, 2024). CL encourages learners to analyze, question, critique, and transform the information they encounter, giving them the ability to foster deep and nuanced understandings of the sociological, cultural, political, and ethical aspects of texts and contexts (Freire, 1970). It is more expansive than ethical reasoning, which concentrates on moral evaluation and decision-making (Bajovic & Elliott, 2011; Luke, 2018). Many theoretical articles (Luke, 2013) and teacher-authored accounts (Vasquez, 2014) detail CL practices in K-12 classrooms. Lewison, Flint, and Sluys (2002) reviewed 30 years of CL literature, synthesizing it into four interrelated dimensions: (1) disrupting the commonplace, (2) considering multiple viewpoints, (3)

Abbreviations: AI, Artificial Intelligence; CL, Critical Literacy; GAN, Generative Adversarial Network; ML, Machine Learning

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focusing on the sociopolitical, and 4) taking action. The model, rooted in sociocultural perspectives, is framed so learning objectives within the dimensions can be adapted across various contexts (Pangrazio, 2016).

Despite the scarcity of learning activities focusing on the wider implications of AI in formal K-12 curricula, significant efforts have been made to develop and study these types of learning activities in formal and informal settings (Smith et al., 2023).

To understand how CL of AI has been supported in children aged 5 to 18, we surveyed the literature to identify records detailing activities supporting young people's reflection on AI's social, political, cultural, ecological, and ethical impacts. Henceforth, we will refer to these kinds of impacts as *critical implications*. First, we conducted a scoping review of publications from the last 10 years to determine: (a) 'What is the current state of the literature?', (b) 'Who has been involved in young people's learning about the critical implications of AI?', (c) 'What has been addressed regarding the critical implications of AI?', (d) 'How has this been addressed?'. Second, we employed a 'best fit' framework synthesis (Carroll, Booth, Leaviss, & Rick, 2013) to scrutinize how Lewison et al.'s (2002) CL learning objectives have been addressed in the literature. This approach also facilitated the synthesis of new learning objectives, grounded in analyzed data, for what we will call *critical AI literacy*. As such, we begin addressing the question: (e) 'What is critical AI literacy, and what can it look like?'. Our aim was not to craft a prescriptive set of CL standards or benchmarks for K-12 AI education, but to inspire further research on supporting learners' critical dispositions toward AI. Therefore, we provide an overview of current implementations of several curriculum design elements and identify potential gaps to explore.

2. Background

2.1. A brief history of critical literacy

In his seminal work, *Pedagogy of the Oppressed*, Paulo Freire (1970) defined *critical literacy* as the ability to analyze, critique, and transform social, cultural, and political texts and contexts. Rooted in language arts, he argued traditional schooling's literacy followed a 'banking model' of education, disregarding learners' lives, cultures, knowledge, and aspirations. He advocated for a dialogical approach to literacy based on ethical principles of reciprocal exchange and recursive processes of discovery.

While critical pedagogy may manifest in a variety of ways (Giroux, 1992; Hooks, 1994; McLaren, 2015), the goal is for students to develop a critical awareness of societal structures of oppression and actively challenge them. Based on Marxist philosophy, Freire emphasizes the development of criticality also requires *Praxis*, defined as "*the action and reflection ... upon the world in order to transform it*" (Freire, 1970, p. 52). Consequently, his work denotes the importance of not only studying the world but also highlighting students' responsibility to create a more just reality.

In the current landscape of modern technocapitalism and the proliferation of unexplainable AI (Ali et al., 2023), Freire's conceptualization of CL may act as a pertinent lens to guide AI literacy education. However, while Freirian paradigms provide educators with a promising alternative to technocratic teaching approaches, there is ambiguity about how teachers and students can concretely engage in CL and consequent Praxis.

2.2. Critical literacy dimensions

In response to Freire's ambiguity, Lewison et al. (2002) posed the questions "*what is critical literacy?*" and "*what does it look like in the classroom?*". They reviewed the literature of educators, theorists, and linguists following the 30 years after Freire wrote his seminal book. Along with their work with elementary school teachers on CL, they

synthesized their understandings into learning objectives under four interlinked dimensions:

Disrupting the commonplace focuses on seeing the everyday through new lenses. The objective is to critique the 'now' and how this 'now' is portrayed in media. To achieve this, children should understand knowledge is constructed throughout time (Shor, 1987) and develop the language to critique and envision better futures (Shannon, 1995). This dimension also calls for using critical self-reflection by studying how language shapes identity and constructs cultural discourses (Fairclough, 1989). In doing so, children are enabled to question their personal assumptions about the world and interrogate the ways in which language may act as a vehicle for shaping the various intersections of their personal and social identities and their corresponding relationships to dominant ideologies.

Considering multiple viewpoints asks us to understand experience and texts from others' viewpoints as well as ourselves, and to consider multiple perspectives concurrently. This includes investigating whose voices are being heard and whose are missing (Luke & Freebody, 1997), examining competing narratives (Farrell, 1999), and making differences visible (Harste, 2000). This dimension provides learners with more holistic insights, but lacks criteria for students to evaluate preferred perspectives.

Focusing on the sociopolitical draws the learners' attention on how sociopolitical systems, power relationships, and language are intertwined. The primary objectives of this dimension are to foster learners' awareness of the complex sociopolitical system to which they belong, to challenge power relationships, and to engage in everyday politics.

Taking action has the objective to use literacy to achieve social justice and is closely related to Freire (1970)'s notion of Praxis. In translating theoretical understandings into practical initiatives within their community, children can be encouraged to envision how social action can change existing discourses (Vasquez, 2014).

It should be noted that these four dimensions of CL are not necessarily exhaustive, nor are the various learning objectives and examples outlined in the article that introduced the dimensions (Lewison et al., 2002). And while Lewison et al.'s (2002) framework gives a good overview of how CL has been conceptualized and instantiated since Freire (1970)'s seminal work, it remains grounded on traditional, language arts, definitions of literacy. Questions have been raised regarding the applicability of these types of frameworks for analyzing contexts beyond text and media (Coiro, Knobel, Lankshear, & Leu, 2014).

2.3. Critical digital literacies

The development of multimedia and emergent technologies has challenged traditional definitions of CL. Media is now more than written text — the multimodality of content and interactive digital contexts suggests the need for new forms of CLs. Vee (2017) argues computer programming augments literacy as an extension of communication, and that literacy studies are "*necessarily interdisciplinary*" (p. 41). Similarly, Lee and Soep's (2016) critical computational literacy framework combines CL with computational thinking in recognition of the confluence of consumer and producer roles. *Consumers* are described by Monroy-Hernández and Resnick (2008) as individuals who view and interact with others' works, whereas *producers* provide feedback and create content. Since Lewison et al.'s (2002) CL framework is based on sociocultural perspectives of literacy, it can be translated across contexts, media, and technologies (Pangrazio, 2016).

Proposing a post-human CL for AI, Leander and Burris (2020) describes CL frameworks as mediating between readers, writers, and social actions. The introduction of AI adds a nonhuman element to this relationship and instigates the evolution of both consumption and production. The authors recommend expanding critical inquiry to pose new questions such as "*Who built the computational agent(s), why and*

how do they operate?” or “How does machine vision, machine text generation and machine text distribution create political influence?” (Leander & Burris, 2020, p. 1273–1274). For example, by examining the impact of deepfakes, children may develop a more nuanced understanding of misinformation and its spread. Likewise, exploring content created by generative adversarial networks (GANs) can draw attention to issues of algorithmic bias. We propose extending criticality towards action by expanding upon conventional applications of critical digital literacies to include emerging AI technologies. Hence, we build upon the aforementioned CL dimensions of Lewison et al. (2002) in the context of AI.

By constructing their own AI futures, children can materialize their critical action (Pangrazio, 2016). However, through an examination of prior works, Ng, Leung, Chu, and Qiao (2021) found that most AI literacy studies focused on lower-level thinking activities according to Bloom’s taxonomy (Bloom, Englehart, Furst, Hill, & Krathwohl, 1956), such as general knowledge and understanding. In contrast, our review puts emphasis on critical AI literacy involving higher-level thinking activities from the revised Bloom’s taxonomy (Anderson & Krathwohl, 2001): analyzing, evaluating, and creating. This taxonomy has been posited as a metacognitive framework to support critical thinking skills (Athanasios, McNett, & Harvey, 2003). By engaging in higher-level activities, students can think beyond conventional paradigms and provide personal critiques and creations (Burbules & Berk, 1999; Tabrizi & Rideout, 2017). Therefore, we present our critical AI literacy framework as learning objectives organized by the revised Bloom’s taxonomy framed within Lewison et al. (2002)’s CL dimensions.

3. Method

This review aimed to gather research papers from the last 10 years that actively support children’s CL regarding AI technologies to examine the extent and nature of this research and synthesize findings into a framework of learning objectives for critical AI literacy. The scoping review methodology used in this study is based on Arksey and O’Malley (2005) approach and its reporting on the PRISMA Extension for Scoping Reviews (Tricco et al., 2018). We use a ‘best fit’ framework synthesis method to synthesize our findings, utilizing a priori codes derived from the literature and the creation of new codes where necessary (Carroll et al., 2013; Ritchie & Spencer, 1994). Our a priori codes are based upon Lewison et al.’s (2002) CL dimensions.

3.1. Search strategy

Five databases were searched: the ACM Digital Library, IEEEExplore, Scopus, ERIC, and PsycINFO. We searched titles, abstracts, and author keywords using a search string combining multiple terms related to study population characteristics, AI technologies, and CL methods (see Table 1 for a complete list). The search was concluded on November 11th, 2023 and yielded 1179 records which were imported into the Covidence review management application. We additionally searched the repository of the International Society of Learning Sciences and Google Scholar with a simplified set of keywords, through which we identified 19 studies that were added to our review database. After eligibility assessment, we conducted a snowball search with a forward search (Wohlin, 2014) of the reference lists of the retained articles. An additional 63 relevant records were retrieved. For an overview of the search, see Fig. 1.

3.2. Selection strategy

Articles were selected for review if they met the eligibility criteria. To be considered for inclusion, records must report on empirical research focused on children’s CL regarding AI technology. To clarify, the learning activities within the study should aim to enable children’s

Table 1

Search strategy keywords. Keywords in each category were separated by the boolean operator “OR”. Each category was separated by the boolean operator “AND”.

Category	Search keywords
<i>Population</i>	Child, Children, Youth, Adolescent, Adolescence, Teen, Preteen, Girls, Boys
<i>AI technology</i>	AI, Artificial Intelligence, Machine Learning, Deep Learning, Child-Agent Interaction, Conversational Agent, AI Literacy, AI ethics, Generative AI, Generative Artificial Intelligence, ChatGPT, Large Language Models, LLM, Robot, Smart Toy, Automation, Child-Robot Interaction
<i>Critical literacy</i>	Ethic*, Critical Literacy, Critical Pedagogy, Critical Inquiry, Critical Making, Computational Empowerment, Participatory Design, PD, Co-design, Perceptions of fairness, Data Ethics, Cooperative Inquiry, Speculative Design, Design Fiction, Design-based Learning, Reimagination, Co-speculation, Critical Design, Diversity and Inclusion, Representational Harm, Perceive

* Indicates truncation.

Table 2

Record selection — Inclusion and exclusion criteria for title/abstract screening and full-text eligibility analysis.

	Inclusion criteria	Exclusion criteria
<i>Phenomenon</i>	Records concerning children’s critical literacy of AI technology	Records concerning technical skill learning, user-testing of AI technology, co-design of AI products without reflection on ethical implications, children’s ontological beliefs of AI technologies
<i>Population</i>	Children between the ages of 5 to 18	Adult participants
<i>Reporting</i>	The most recent, peer-reviewed, English records on a study	Research protocols, review, editorials, book chapters, magazines, books and other grey literature

analysis, critique, or transformation of an AI technology’s critical implications (i.e., social, political, cultural, ecological, and ethical impacts). We focused on studies and activities that had explicit supports for critical engagement rather than those where critical reflection emerged spontaneously. Papers in which critical reflections naturally emerged were excluded. The study should involve children between the ages of 5 and 18. We only included English peer-reviewed publications from the last 10 years, excluding non-empirical, grey literature, and non-English papers. We also excluded studies reporting on children’s perceptions of an AI’s human-likeness, children’s preconceptions of AI, children’s opinions of AI systems without support for critical reflection, children’s ability to detect AI technologies or AI-created media, children user-testing an implementation of an AI system, and children co-designing AI systems for specific problems, as here the focus is on the implementation domain and not on the critical implications of technology. In cases where we identified multiple records concerning the same study, only the most recent publication was included. A list of the inclusion and exclusion criteria can be found in Table 2.

Titles and abstracts were screened by two researchers for mentions of the inclusion or exclusion criteria. At this stage, 179 studies were retained with almost perfect agreement ($r = 0.93$; $\kappa = 0.865$) (McHugh, 2012). After retrieval, 178 full texts were assessed for eligibility by two researchers. The agreement on study inclusion was substantial ($r = 0.92$; $\kappa = 0.705$). Disagreements were always settled through discussion between the first, second, and third authors. A total of 30 records were retained.

3.3. Data extraction and analysis

In accordance with the questions framed in the introduction, we started with extracting data regarding four categories: (a) descriptive

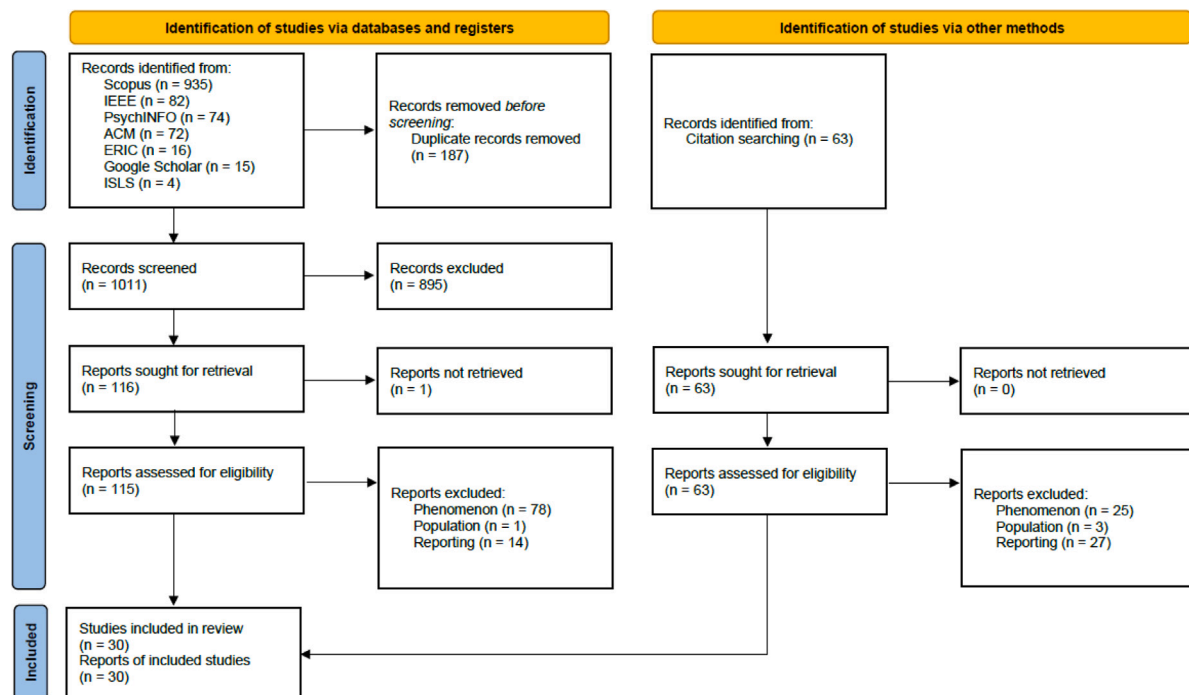


Fig. 1. PRISMA flow diagram of search, screening, and selection of evidence records.

information, b) participants, c) technology, and d) learning activities. Our focus was specifically on the AI technologies children had critically analyzed and the activities they engaged in to do so. We then delved deeper into these learning activities to analyze the critical AI literacy learning objectives within. Since much of this work was interpretive, all extraction and analysis was carried out independently by at least two researchers before reaching consensus on a final decision.

All authors deliberated on the data extraction protocol. Each extraction target was accompanied by a leading question and example to ensure all authors were on the same page. Extracted descriptive information included the title, author information, year of publication, publication venue, publication type, and geographical location of the study population. Regarding participants, we extracted the number of children involved in the study and their demographic characteristics with a focus on different intersections of identity. Additionally, we documented the adults involved in these studies and identified their roles. The technology category aims to lay bare the content of the children's learning. We extracted the various AI technologies and their critical implications explored by children. Finally, descriptions of all learning activities undertaken by children within the study were gathered from the methods sections. We then separated activities that allowed children to analyze, critique, or transform AI's critical implications for further analysis.

The extracted AI technologies were organized into one of three categories adhering to Long and Magerko's (2020) classification of AI systems. Then, critical implications were categorized based on a priori categories defined by Long and Magerko (2020). Critical implications outside of these categories were classified inductively. The learning activities were grouped by learning theory, then subdivided into pedagogical approaches inspired by the process and categories used by Yue, Jong, and Dai (2022). To determine the CL learning objectives represented in these learning activities, we conducted framework synthesis of the learning activities data with a "best-fit" approach (Carroll et al., 2013) using Lewison et al. (2002)'s CL framework as a starting point. We familiarized ourselves with the framework and adapted the CL learning objectives to the context of AI by either renaming terms

pertaining to language arts-related terms to 'AI' – for instance, study the relationship between *language* and power became to study the relationship between *AI* and power – or adding '*regarding AI*'. Then, we classified the new learning objectives according to the levels of the revised Bloom taxonomy (Anderson & Krathwohl, 2001), and composed a description for each objective (see A for an overview). This served as a priori codes against which the extracted activity data was mapped. New learning objectives were created using an inductive approach when data did not match existing codes. Each proposed learning objective was discussed and agreed upon for interpretation, data support, and alignment with preexisting codes before being added. Thus, a revised conceptual framework built upon Lewison et al.'s (2002) was constructed to describe learning objectives for critical AI literacy based on the data.

3.4. Limitations

First, this review only covers archival journal and conference proceedings records. A large number of grey literature publications have been published on the subject of critical AI literacy (e.g., DiPaola, 2021; Druga, 2018; Van Brummelen, 2022). However, to ensure academic quality of our data, we only included peer-reviewed records. Second, when deciding to exclude and include papers, we did not consider developmental requirements to participate in CL activities.

We do not discuss prerequisite knowledge or fundamental capacities needed for children to engage in critical AI literacy. Third, studies in our corpus include both research and learning activities. In other words, some studies were designed to involve children as informants (Druin, 2002), whereas others describe learning trajectories. Rather than focusing on study outcomes, however, we were interested in the potential for learning activities to foster critical AI literacy, enabling us to review both types of articles. The fourth limitation relates to the clarity and accessibility of the data for our framework synthesis. We consulted results sections and supplementary documents for contextual clarification of our data. Unfortunately, some materials were unavailable. Consequently, some records may include critical AI literacy learning objectives that we could not identify.

Table 3

Summary of descriptive information of selected studies.

Study	Venue	Country	Formal	Remote	Study design	Participants	Age
Aki Tamashiro, Van Mechelen, Schaper, and Sejer Iversen (2021)	IDC†	Denmark	N/A	Y	Case study	$N(c) = 2$	14–15
Ali et al. (2021)	Journal	USA	N	Y	Case study	$N(c) = 38$, $N(i) = 4$	10–15
Arada, Sanchez, and Bell (2023)	Journal	USA	Y	N	Case study	$N(c) = 2$ ♣, $N(i) = 2$	13–14
Arastoopour Irgens, Adisa, Bailey, and Vega Quesada (2022)	Journal	USA	N	N	Participatory design	$N(c) = 44$, $N(i) = 3$	9–13
Bilstrup, Kaspersen, and Petersen (2020)	DIS	Denmark	Y	N	Constructive design research	$N(c) = 66$, $N(i) = 4$	16–20
Buddemeyer et al. (2022)	GoodIT†	USA	N	N	Participatory design	$N(c) = 12$ ♣	11–13
Charisi, Imai, Rinta, Nakhayenze, and Gomez (2021)	IDC†	Uganda and Japan	Y	N	Participatory action research	$N(c) = 44$	6–16
Dai, Lin, Liu, Dai, and Wang (2023)	Journal	Hong Kong SAR	Y	N	Quasi experimental	$N(c) = 83$, $N(i) = 1$	11–12
DiPaola, Payne, and Breazeal (2020)	IDC	USA	N	N	Case study	$N(c) = 19$	10–14
Everson, Kivuva, and Ko (2022)	SIGCSE	USA	N	Y	Constructive design research	$N(c) = 14$ ♣	15–17
Forsyth et al. (2021)	RESPECT†	USA	Y	Y	Case study	$N(c) = 73$, $N(i) = 3$	12–18
Harrington and Dillahunt (2021)	CHI	USA	N	Y	Co-design	$N(c) = 5$ ♣	16–19
Kaspersen et al. (2022)	Journal	Denmark	Y	N	Constructive design research	$N(c) = 61$, $N(i) = 2$	17–20
Krakowski, Greenwald, Hurt, Nonnecke, and Cannady (2022)	AAAI	USA	Y	Y	Case study	$N(c) = 24$	15–18
Lee, Ali, Zhang, DiPaola, and Breazeal (2021)	SIGCSE	USA	N	Y	Case study	$N(c) = 31$ ♣	10–15
Lee, Gobir, Gurn, and Soep (2022)	Journal	USA	N	N	Participatory design	$N(c) = 16$ ♣, $N(i) = 4$	15–19
Lin, Van Brummelen, Lukin, Williams, and Breazeal (2020)	AAAI	USA	Y	N	Case study	$N(c) = 14$	8–11
Lin et al. (2023)	Journal	China	Y	N	Experiment	$N(c) = 79$, $N(i) = 1$	11–12
Mott, Bejarano, and Williams (2022)	HRI	USA	Y	N	Co-design	$N(c) = 20$	7–11
Ng, Luo, Chan, and Chu (2022)	Journal	Hong Kong SAR	Y	N	Case study	$N(c) = 82$, $N(a) = 16$	6–12
Rankin and Henderson (2021)	Journal	USA	N	N	Case study	$N(c) = 10$ ♣, $N(i) = 2$	9–18
Salac, Landesman, Druga, and Ko (2023)	IDC	USA	N	Y	Case study	$N(c) = 16$	8–12
Schaper, Malinverni, and Valero (2020)	NordiCHI†	Spain	Y	N	Case study	$N(c) = 23$ ♣, $N(i) = 1$	11–12
Schaper et al. (2022)	Journal	Denmark	Y	N	Participatory design	$N(c) = 26$, $N(i) = 9$	14–15
Schaper, Tamashiro, Smith, Van Mechelen, and Iversen (2023)	IDC	Denmark	Y	N	Participatory design	$N(c) = 11$	14–15
Skinner, Brown, and Walsh (2020)	CHI†	USA	N	N	Co-design	N/A	9–14
Solyst, Axon, Stewart, Eslami, and Ogan (2022)	ICLS	USA	N	Y	Case study	$N(c) = 32$	11–14
Solyst, Xie et al. (2023)	CHI	USA	N	N	Case study	$N(c) = 10$ ♣, $N(i) = 2$	9–11
Solyst, Yang et al. (2023)	Journal	USA	N	N	Case study	$N(c) = 30$ ♣, $N(i) = 1$, $N(a) = 6$	6–18
Williams et al. (2023)	Journal	USA	N	Y	Case study	$N(c) = 78$, $N(i) = 11$	11–13

Annotation. *Venue*: the (†) indicates short or work-in-progress conference articles. *Participants*: $N(c)$ are the number of children who participated in the study, $N(i)$ refers to the number of instructors that were involved in the activity other than the researchers themselves and $N(a)$ refers to the number of adults that participated in the study. The (♣) indicates studies that purposefully recruited participants from minority groups.

4. Results

We report our results in two main sections. First, we present data from our scoping review, including the corpus' descriptive information, and participants, technology, and learning activities. We then present our framework synthesis of critical AI literacy learning objectives. An overview of the 30 reviewed studies and their descriptive characteristics is presented in Table 3.

4.1. Descriptive information

Despite our search covering a decade of research, the records in this review have an inclusive publication range from 2020 to 2023. Five

records were published in 2020, seven in 2021, eleven in 2022, and seven in 2023 (see Fig. 2). A majority were published in conference proceedings ($N = 18$), of which twelve were full conference articles and six were extended abstracts or short papers. The included records have been published in a wide variety of conference proceedings (see Table 3). The most common conference venues included the ACM Interaction Design and Children conference (IDC, $N = 5$), and the conference on Human Factors in Computing Systems (CHI, $N = 3$). The remaining twelve articles were published as full journal papers.

A majority of the studies in this sample have been conducted in North America, with 20 studies conducted in the United States. In Europe, five studies occurred in Denmark and one in Spain. Four studies took place in Asia—two in Hong Kong SAR and one in China.

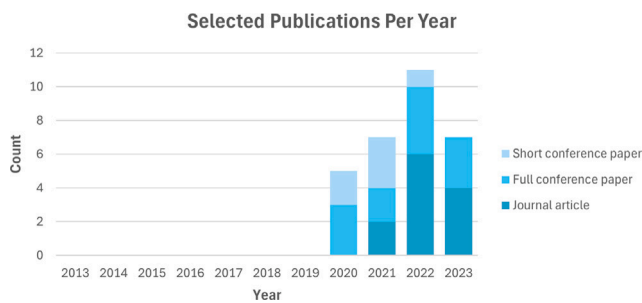


Fig. 2. Selected 30 publications are published from 2020–2023 inclusively.

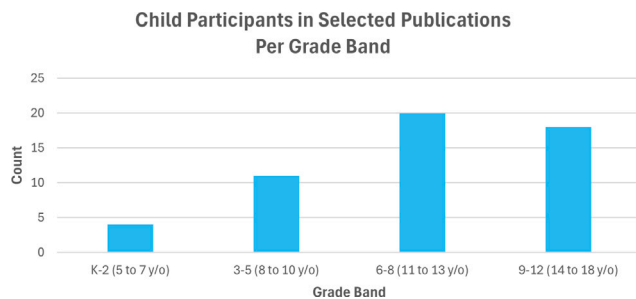


Fig. 3. Child participants in selected publications per grade bands.

Additionally, one study, conducted by Charisi et al. (2021), took place in both Japan and Uganda. Fourteen studies took place in formal educational settings, while fifteen were independent of school curricula but focused on getting children thinking about AI's critical implications in informal learning environments.

Additionally, 33% of the studies within this sample ($N = 10$) were conducted remotely.

Regarding study design, sixteen papers reported a form of case study design. Participatory methods were used through participatory design projects ($N = 5$), co-design studies ($N = 3$) and participatory action research ($N = 1$). Three papers reported iterative development of their learning activities in constructive design research studies. Lastly, two studies investigated the effectiveness of their activities through quasi-experimental ($N = 1$) or true experimental ($N = 1$) designs.

4.2. Participants and their intersections

The sample size of the included studies ranged from 2 to 83 child participants. One study did not indicate the total number of children involved (Skinner et al., 2020). The ages of the child participants encompassed a spectrum ranging from 6 to 20 years old (see Table 3). To determine frequency, we sorted participants in grade bands congruent with Touretzky et al. (2019)'s banding for AI K-12 literacy guidelines. The two articles including students older than 18 were sorted in the grade 9–12 band (Bilstrup et al., 2020; Kaspersen et al., 2022).

As can be seen in Fig. 3, most studies included children with ages corresponding to higher grade bands, with a majority of studies including children between the ages 11 to 13 ($N = 20$) and 14 to 18 years old ($N = 18$). Fewer articles involved younger children between the ages of 8 to 10 years ($N = 11$) and 5 to 7 years ($N = 3$). Furthermore, half of the publications involved participants from multiple grade bands ($N = 15$).

With respect to intersectionality, several papers mention gender and racial inequities within STEM learning. While most studies included participants from mixed genders ($N = 18$), eight papers exclusively involved children identifying as a gender minority in STEM (girls and non-binary individuals). 15 articles involved children belonging to racial and ethnic minorities in STEM learning. Of those papers, a

significant subset ($N = 10$) explicitly prioritized Black, Indigenous, and People of Color (BIPOC) representation through their recruitment process (see Table 3).

Beyond intentionally recruiting participants of color, five papers used a Black feminist lens to guide the design of learning activities and study (Arada et al., 2023; Buddemeyer et al., 2022; Harrington & Dillahunt, 2021; Rankin & Henderson, 2021; Solyst, Xie et al., 2023). Only one paper reported the involvement of a child with ADHD (Salac et al., 2023) and none mentioned the inclusion of children with physical disabilities.

Participant socioeconomic status was generally undisclosed. One paper classified its sample from a socioeconomically disadvantaged area (Skinner et al., 2020) and another from a socioeconomically advantaged area (Schaper et al., 2023).

Aside from child participants, we also identified how adult participants were involved. Despite being integral to the execution of learning activities, educators are often unfamiliar with implications of emerging technologies such as AI (Stoiber, Kim, & Kim, 2023). Therefore, there have been calls within CCI research for more sustainable “participatory practices” (Smith et al., 2023). We identified eleven studies where adults co-facilitated learning activities. Less than half of the studies reported providing professional development training and/or tools to support co-facilitators in executing activities (Arada et al., 2023; Charisi et al., 2021; Dai et al., 2023; Williams et al., 2023). Additionally, three records reported involving both adults and children as participants, either through collaboration (Ng et al., 2022; Skinner et al., 2020) or parallel practice (Solyst, Yang et al., 2023).

4.3. AI technologies and their critical implications

The AI technologies were extracted only from activities that led children to consider their critical implications. They were organized into one of three categories (Long & Magerko, 2020).

Cognitive systems are based on theories regarding human cognition, encompassing topics including knowledge representations, decision-making, problem-solving, and learning. *Machine learning* (ML) is based on algorithms and statistics. Though similar, the two aforementioned categories differ: a cognitive system can produce varied outputs for the same prompt, while ML should consistently generate the same output for identical prompts. *Robotics* revolves around AI systems capable of physical actions.

We used a set of categories for AI ethics defined by Long and Magerko (2020) to categorized the critical implications of these AI technologies. These included accountability, algorithmic bias, diversity in the workforce, employment, moral decision-making, misinformation, privacy, surveillance, and transparency. Moreover, nine implications emerged inductively: (1) *Harms* refer to the broad range of potential consequences ideated by children; (2) *Identity representation* relates to the portrayal of diverse backgrounds through AI technologies; (3) *Authority* involves power relationships such as in child–robot interactions; (4) *Democratic design* includes traditionally marginalized voices in the design of AI systems; (5) *Civics* entails the rights and duties of a citizen; (6) *Fairness* is a broader consideration of justice in the context of AI without an algorithmic focus; (7) *Environment* includes impacts on the planet; (8) *Ownership* refers to the proprietary rights of AI-generated content, systems, and data; and (9) *Reliance on AI* suggest repercussions from overdependence on these systems. Fig. 4 provides an overview of the frequency of each identified critical implication.

4.3.1. Cognitive systems

As summarized in Table 4, ten publications investigated cognitive systems, which includes intelligent agents, Generative Adversarial Networks (GANs), and autonomous vehicles. Application contexts were kept general to daily life or society at large. Studies focusing on intelligent agents ($N = 5$) included discussions about algorithmic bias, identity representation, and fairness. Buddemeyer et al. (2022), for

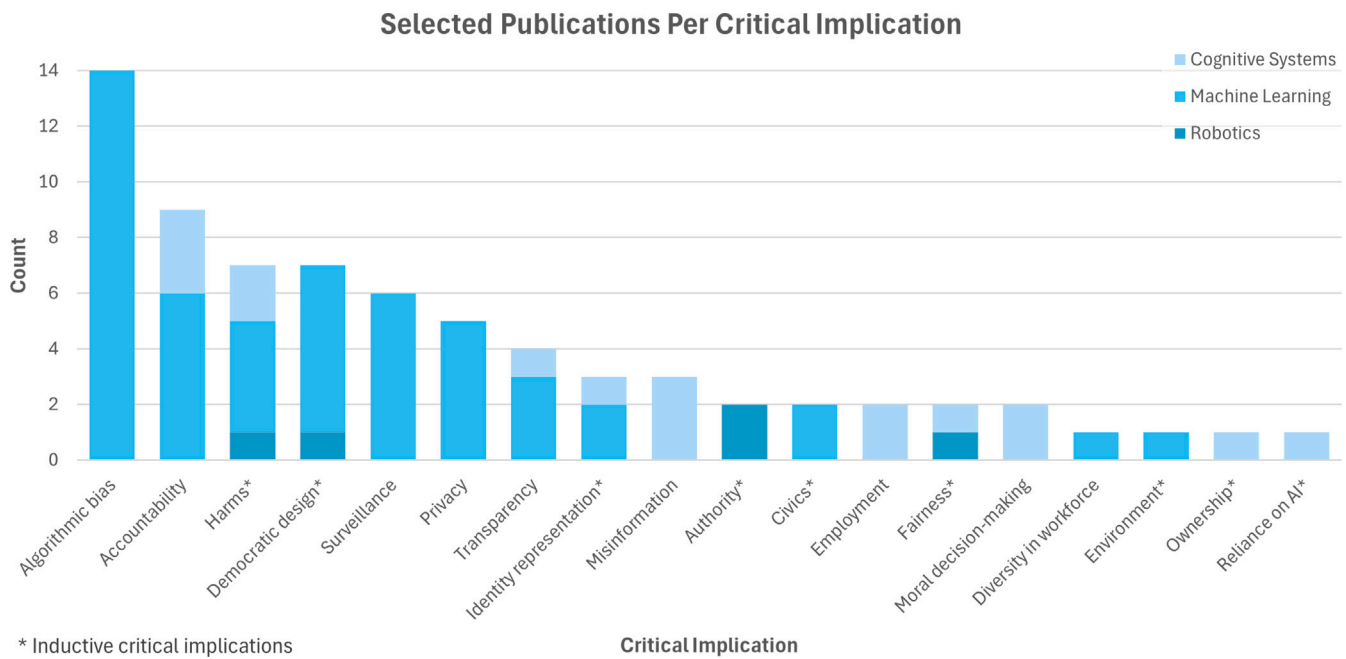


Fig. 4. Selected publications organized by category of AI technology per extracted critical implication. This graph does not include a count for studies with activities supporting design opportunities. An asterisk (*) denotes an implication that emerged inductively. Note that [Schaper et al. \(2023\)](#) is not included as the discussion of harms is vaguely general across various technologies and children suggested their own ideas. [Dai et al. \(2023\)](#) is excluded from the graph as the activities discuss ethics in a broad sense. [Solyst, Xie et al. \(2023\)](#) also includes considerations of “fairness” both unspecific to AI and vaguely within the context of AI.

instance, recruited Black girls to examine the design of AI dialogue, leading to additional considerations about how language may differ based on identity and representation of marginalized voices in conversational agents.

In other studies, children considered more general benefits and harms of cognitive system deployments to society ([Dai et al., 2023](#); [Lin et al., 2020](#); [Skinner et al., 2020](#); [Solyst et al., 2022](#)), as well as wider reliance of society on these systems ([Lin et al., 2020](#)).

We distinguish between deepfakes that insinuate misinformation as content, and general image generation using GANs because the resulting issues differ. Three studies stressed the topic of misinformation and divulged techniques to identify deepfakes ([Ali et al., 2021](#); [Lee et al., 2021](#); [Williams et al., 2023](#)). Conversely, ([Lee et al., 2021](#)) incorporated a discussion about art ownership.

Three studies encouraged critical reflection on autonomous vehicles, two of which involved moral decision-making and accountability ([Forsyth et al., 2021](#); [Lin et al., 2023](#)) through a trolley-problem. Additionally, two papers use AI-assisted vehicles as a springboard to explore the consequential effects on future careers ([Lee et al., 2021](#); [Lin et al., 2023](#)), such as improving efficiency, creating new occupations, or taking over existing jobs. Furthermore, [Forsyth et al. \(2021\)](#) bolsters transparency by disclosing how AI works and uses data, enabling children to participate more critically.

4.3.2. Machine learning

Twenty-one papers discussed ML, primarily within the context of daily life. However, six papers also included critical discussions relating to the context of entertainment such as recommendations for music and videos ([DiPaola et al., 2020](#); [Lee et al., 2022](#); [Solyst, Xie et al., 2023](#); [Williams et al., 2023](#)), game NPCs ([Dai et al., 2023](#)), or face filters creation ([Aki Tamashiro et al., 2021](#)). The extracted types of ML technologies are classification, facial recognition, recommender systems, and voice recognition.

The most prevalent critical implication involved algorithmic bias ($N = 14$), which covers issues like stereotyping and voice or facial misrecognition. Youth and children learned about how historical biases ([Salac et al., 2023](#); [Solyst, Xie et al., 2023](#)) and insufficiently

Table 4

Overview of the AI technologies discussed within the included records sorted by categories defined by [Long and Magerko \(2020\)](#).

Category	AI technology
<i>Cognitive systems</i> ($N = 10$)	Intelligent agents (Buddemeyer et al., 2022 ; Dai et al., 2023 ; Lin et al., 2020 ; Skinner et al., 2020 ; Solyst et al., 2022) GANs: Deepfakes (Ali et al., 2021 ; Lee et al., 2021 ; Williams et al., 2023) GANs: Images (Lee et al., 2021) Autonomous vehicles (Forsyth et al., 2021 ; Lee et al., 2021 ; Lin et al., 2023)
<i>Machine learning</i> ($N = 21$)	Classification (Arada et al., 2023 ; Arastoopour Irgens et al., 2022 ; Bilstrup et al., 2020 ; Dai et al., 2023 ; Everson et al., 2022 ; Forsyth et al., 2021 ; Kaspersen et al., 2022 ; Krakowski et al., 2022 ; Ng et al., 2022 ; Salac et al., 2023 ; Schaper et al., 2022 ; Solyst et al., 2022 ; Solyst, Xie et al., 2023 ; Solyst, Yang et al., 2023 ; Williams et al., 2023) Facial recognition (Aki Tamashiro et al., 2021 ; Forsyth et al., 2021 ; Harrington & Dillahunt, 2021 ; Lee et al., 2022 ; Schaper et al., 2022 ; Solyst, Xie et al., 2023) Recommender systems (DiPaola et al., 2020 ; Forsyth et al., 2021 ; Lee et al., 2021, 2022 ; Salac et al., 2023 ; Solyst, Xie et al., 2023) Voice recognition (Rankin & Henderson, 2021 ; Salac et al., 2023 ; Solyst, Xie et al., 2023 ; Williams et al., 2023)
<i>Robotics</i> ($N = 4$)	Androids (Charisi et al., 2021 ; Mott et al., 2022 ; Schaper et al., 2020) Robots (Solyst et al., 2022)

Note: [Schaper et al. \(2023\)](#) is not included in this table because activities did not focus on a specific type of intelligent technology.

diverse datasets ([Arada et al., 2023](#); [Arastoopour Irgens et al., 2022](#); [Bilstrup et al., 2020](#); [Dai et al., 2023](#); [Everson et al., 2022](#); [Forsyth et al., 2021](#); [Kaspersen et al., 2022](#); [Krakowski et al., 2022](#); [Ng et al., 2022](#); [Schaper et al., 2022](#); [Solyst et al., 2022](#); [Solyst, Yang et al., 2023](#); [Williams et al., 2023](#)) can perpetuate discrimination in AI classification and decision-making. For example, if systematic issues of racism are reflected in statistical data; it can become problematic if prejudiced views are reinforced by AI systems such as with predictive policing and biased search results.

Children contemplated how AI systems learning from these datasets can better represent intersectional populations and themselves (Harrington & Dillahun, 2021; Rankin & Henderson, 2021).

Moreover, critical discussions centered around data introduced concerns around democratic design ($N = 6$), accountability ($N = 6$), surveillance ($N = 6$), privacy ($N = 5$), and transparency ($N = 3$). All six studies with relevant activities focusing on facial recognition include discussions regarding surveillance. While surveillance focuses primarily on organizational or governmental monitoring, privacy can also be about the sharing of personal data with companies (Schaper et al., 2022). These matters are interrelated with transparency of the system; it is not always clear what the company is doing with data collected from its users (DiPaola et al., 2020). Some studies support discussions of transparency by demystifying how AI systems work (Bilstrup et al., 2020; Forsyth et al., 2021; Lee et al., 2022), such as demonstrating the hidden metrics behind recommender systems.

A few papers produced additional unique critical considerations. Bilstrup et al. (2020) and Kaspersen et al. (2022) present the case of using ML to predict voter decisions in democratic elections, facilitating discussions around possible consequences in a political context. Furthermore, Salac et al. (2023) raise concerns about diversity in the AI and computer science workforce, and Williams et al. (2023) discuss environmental impacts of large datasets.

4.3.3. Robotics

A total of four papers discussed robotics, three of which were about androids (Charisi et al., 2021; Mott et al., 2022; Schaper et al., 2020), which are humanoid robots. In both Mott et al. (2022) and Schaper et al. (2020), children considered how a physical AI technology could hold authority in the contexts of a social robot (e.g., caregivers, teachers, companions) and a robot president, respectively. By deliberating upon human–robot interactions, children additionally considered the impacts of the system in wider perspectives such as whether decisions made by robots are fair (Charisi et al., 2021) and how they may be beneficial or harmful (Solyst et al., 2022). Additionally, Schaper et al. (2020) also supported children to consider whether a robot could understand or have cultural values, and whether this could make a robot president a better representative of the people it governs and more inclusive to diverse voices in the community.

4.4. Pedagogical approaches, learning activities, and materials

The **constructivist** learning theory is commonly adopted in AI literacy education (Yue et al., 2022). It posits that students construct the meaning of knowledge from experience and stresses the key role of sociocultural factors in students’ learning process (Fosnot, 1996). Three constructivist pedagogical approaches were identified: *dialogic learning* ($N = 38$), *interactive learning* ($N = 10$), and *inquiry-based learning* ($N = 4$). Where constructivist learning emphasizes the importance of reflection and abstract reasoning, **constructionist** learning theory expands this by emphasizing the importance of tinkering and making in the learning process (Ackermann, 2001). Three articles specifically mention constructionism as a guiding theory (Bilstrup et al., 2020; Lin et al., 2020; Ng et al., 2022). We distinguish between two constructionist pedagogical approaches: *maker education* ($N = 7$) and *design-based learning* ($N = 13$) (see Table 5).

4.4.1. Dialogic learning

Dialogic learning emphasizes dialogue and interaction among participants, facilitating the co-construction of knowledge through dialogue. It encourages critical reflection by fostering a collaborative learning environment where diverse perspectives are shared and valued (Alexander, 2008). We identified six types of activities that can be classified under dialogic learning, four of which were different types of discussions.

Table 5
Overview of learning activities performed in the included records sorted by pedagogical approach.

Pedagogical approach	Learning activity
Constructivism	
<i>Dialogic learning</i> ($N = 38$)	Scenario-based discussion (Charisi et al., 2021; Dai et al., 2023; Forsyth et al., 2021; Lee et al., 2021; Lin et al., 2023; Mott et al., 2022; Schaper et al., 2020; Solyst, Xie et al., 2023; Williams et al., 2023); Layered scenario (Salac et al., 2023; Skinner et al., 2020) Media-informed discussion (Ali et al., 2021; Arada et al., 2023; Arastoopour Irgens et al., 2022; Everson et al., 2022; Harrington & Dillahun, 2021; Lee et al., 2022; Solyst et al., 2022; Solyst, Yang et al., 2023; Williams et al., 2023); Socratic seminar (DiPaola et al., 2020) Reflective discussion (Aki Tamashiro et al., 2021; Bilstrup et al., 2020; Kaspersen et al., 2022; Lin et al., 2020; Schaper et al., 2022) Interaction-informed discussion (Aki Tamashiro et al., 2021; Lee et al., 2022; Rankin & Henderson, 2021) Expert dialogue (Everson et al., 2022) Role-play (Aki Tamashiro et al., 2021; Lin et al., 2023; Skinner et al., 2020)
<i>Interactive learning</i> ($N = 10$)	Simulation (Ali et al., 2021; Arastoopour Irgens et al., 2022; Krakowski et al., 2022; Ng et al., 2022; Williams et al., 2023) ML model construction (Bilstrup et al., 2020; Kaspersen et al., 2022; Krakowski et al., 2022) Moral dilemma (Forsyth et al., 2021; Lin et al., 2023)
<i>Inquiry-based learning</i> ($N = 4$)	Independent research (Everson et al., 2022; Lee et al., 2021) Design requirement (Rankin & Henderson, 2021; Salac et al., 2023)
Constructionism	
<i>Maker education</i> ($N = 7$)	Visionboarding (Arada et al., 2023) Digital story writing (Ng et al., 2022) Multimedia products (Forsyth et al., 2021; Lee et al., 2022) Co-design (Buddemeyer et al., 2022; Mott et al., 2022; Skinner et al., 2020)
<i>Design-based learning</i> ($N = 13$)	Value-sensitive design: Redesign (DiPaola et al., 2020; Lee et al., 2021; Williams et al., 2023); Cultural alternatives (Schaper et al., 2020, 2022) User-centered design (Arastoopour Irgens et al., 2022; Bilstrup et al., 2020; Solyst et al., 2022; Solyst, Xie et al., 2023; Solyst, Yang et al., 2023) Speculative design (Aki Tamashiro et al., 2021; Harrington & Dillahun, 2021; Schaper et al., 2023)

In scenario-based discussion activities ($N = 11$), facilitators provided learners with narratives highlighting critical implications of an AI-based technology, which learners explored through discussions.

Typically, learners were presented with multiple scenarios related to the same topic. At times, adaptations were made to reflect the learners’ cultures (Charisi et al., 2021). Several studies used theoretical approaches to scaffold learners’ reasoning. Dai et al. (2023) guided students’ thinking on the uniqueness of AI through analogical scaffolding (Podolefsky & Finkelstein, 2007). A human–AI analogy was presented along with two cases allowing students to compare and contrast between humans and AI. To support students’ moral development, Lin et al. (2023) used the Konstanz Method of Dilemma Discussion (Lind, 2006) in which educators facilitate a moral debate. Salac et al. (2023) applied sense-making theory (Klein, Phillips, Rall, & Peluso, 2007) by slowly revealing different layers of algorithmic decision-making. At each layer’s unveiling, the learners were prompted to discuss and evaluate choices made by the algorithm based on their understanding of all prior layers.

Other discussions focused on media depicting critical implications of AI. Audiovisual media was the most prevalent. In five records (Arada et al., 2023; Arastoopour Irgens et al., 2022; Everson et al., 2022; Solyst

et al., 2022; Williams et al., 2023), learners were shown videos about algorithmic bias based on the works of Joy Buolamwini and Gebru (2018).

To discuss the implications of deepfakes, students were shown deepfaked videos and images (Ali et al., 2021; Williams et al., 2023). Lastly, some papers used news articles to spark discussion.

For instance, DiPaola et al. (2020) provided learners with an article about children's data privacy regarding YouTube's recommendation algorithm, and engaged them in structured discussing using the Socratic seminar method.

By reflective discussions, we refer to concluding discussions at the end of an activity sequence, which allow students to reflect on their learning and apply it to other situations. In two articles, after learning how ML works, students engage in a discussion about the role of AI in the larger society (Kaspersen et al., 2022; Lin et al., 2020). In other reports, the designs created in prior activities are used as the basis for discussion of ethical and societal implications (Aki Tamashiro et al., 2021; Bilstrup et al., 2020; Schaper et al., 2022).

Interaction-informed discussions are critical dialogues grounded in learner's experiences with an existing AI product. For instance, (Lee et al., 2021) initiated discussion on data privacy by asking youth to write poems using the predictive text feature on their mobile phones.

We also identified two activities using alternative forms of dialogue. In role-play activities, children use dialogue to act as another agent and to understand the actions and motivations of someone else. This can involve assuming the role of different stakeholders (Lin et al., 2023), of AI technology (Skinner et al., 2020) or of agents from the future or different worlds (Aki Tamashiro et al., 2021). Lastly, learners can initiate dialogue by asking questions to an expert. Everson et al. (2022) hosted an AI developer for a guest lecture, enabling students to inquire about critical implications.

4.4.2. Interactive learning

Interactive learning involves students' active hands-on learning, often through the use of technology, to gain an understanding of the critical implications of AI. Activities following this approach include educational technologies that simulate the impact of AI with a predetermined outcome. For example, selected publications employed Google's Teachable Machine¹ with a biased dataset (Arastoopour Irgens et al., 2022), an application simulating the spread of misinformation (Ali et al., 2021; Williams et al., 2023), or the games Survival of the Best Fit² (Krakowski et al., 2022) and AI Ocean³ (Ng et al., 2022).

We identified two types of interactive learning activity where students made critical judgements themselves. ML model construction is a common activity for teaching learners general applicable AI literacy. Authentic scenarios can prompt learners to make critical considerations during their own model development. In Krakowski et al.'s (2022) work, the students selected different features for their ML model and observed how it affected hypothetical college admissions. Bilstrup et al. (2020) and Kaspersen et al. (2022) involved the use of VotestatesML, a tool which allows students to experiment with building models for predicting voter behavior in a civics course. Specifically for autonomous vehicles, researchers used the website Moral Machine⁴ (Forsyth et al., 2021) or an AR-application (Lin et al., 2023) to demonstrate moral dilemmas for learners to judge.

4.4.3. Inquiry-based learning

In inquiry-based learning activities, students conduct independent research on critical implications of AI, such as data privacy (Everson

et al., 2022) or future career opportunities (Lee et al., 2021). They were free to present their findings in any preferred format. As part of inquiry-based learning, we also coded activities where learners consolidate their explorations into design requirements for future AI technologies (Rankin & Henderson, 2021; Salac et al., 2023).

4.4.4. Maker education

The maker education approach involves learners in the development of an artifact with a predefined problem-statement. We identified three types of artifacts where learners externalized their ethical stance regarding AI: visionboards (Arada et al., 2023), digital stories (Ng et al., 2022) and general multimedia products (Forsyth et al., 2021; Lee et al., 2022). For instance, Lee et al. (2022) introduced learners to hidden elements of recommender systems and the surveillance implications of facial recognition technologies. Based on these provocations, youth collaboratively created multi-media artifacts that expressed their critical stance.

Co-design activities permit children a voice in product development, but are often initiated by researchers with specific interests. Therefore, they provide children with a clearly defined problem statement. For instance, Buddemeyer et al. (2022) included marginalized youth in designing conversational patterns for conversational technologies. In this case, researchers determined the type of AI technology, interaction modality, and user group.

4.4.5. Design-based learning

Design-based learning activities provide learners with open-ended project briefs for which they identify their own problem statements through inquiry and solutions through design (Bekker, Bakker, Douma, Van Der Poel, & Scheltenaar, 2015). We identified five design-based learning activities informed by different design paradigms. We found two activities inspired by value sensitive design (VSD) (Friedman & Hendry, 2019). In three records, learners were asked to consider different values that might be held by stakeholders of an existing AI technology, either Youtube (DiPaola et al., 2020; Lee et al., 2021; Williams et al., 2023) or Alexa (Williams et al., 2023). These values spark discussion and new opportunities for redesigning the technology. Two articles combined VSD with cultural considerations. One asked learners to analyze the values of another culture through a case study (Schaper et al., 2022), while the other focused on culturally representative foods to design AI technologies (Schaper et al., 2020).

User-centered design activities prompt learners to envision AI-based technologies for positive change. These activities allowed students to identify problems within their community or daily lives that they might solve with AI (Arastoopour Irgens et al., 2022; Solyst et al., 2022; Solyst, Xie et al., 2023). For evaluation, learners can consider the impact of their AI-based concepts on a potential user's feelings (Bilstrup et al., 2020) or critically evaluate their own designs through prompts such as: *Fair interactions my technology could have are, The people who help with the creation of AI are, They could help by, Unfair interactions I want my technology to avoid are* (Solyst, Xie et al., 2023). Conversely, Solyst, Yang et al. (2023) had participants explore alternative technologies to address algorithmic bias in AI systems.

Rooted in critical origins, speculative activities enable learners to confront and counter their sociotechnical realities through imagining alternative present and future worlds (Bleecker, 2022). For example, by asking learners to collaboratively storyboard narratives about possible utopian and dystopian AI futures (Harrington & Dillahunt, 2021; Schaper et al., 2023) or imagining alien technologies (Aki Tamashiro et al., 2021).

4.5. Learning objectives, taxonomy levels and their qualities

Our analysis concludes by sketching out the beginnings of a framework for critical AI literacy based on learning objectives we identified

¹ <https://teachablemachine.withgoogle.com/>

² <https://www.survivalofthebestfit.com/>

³ <https://code.org/oceans>

⁴ <https://www.moralmachine.net/>

Table 6
Critical AI Literacy learning objectives for the ‘disrupting the commonplace’ dimension.

Disrupting the commonplace	
Taxonomy	Learning objective
<i>Analyze</i>	Question how AI is integrated in their daily life* Question preconceptions of AI and its critical implications* Question “everyday” ways of seeing AI Study AI to analyze how it shapes identity Question how people’s behaviors are constructed by AI technologies Analyze how AI technologies disrupt society’s trajectory and status quo
<i>Evaluate</i>	Judge how AI is applied in daily life* Judge potential applications of AI*
<i>Create</i>	Envision responsible use of AI technologies*

Annotation. *Italicized* words indicate changes that were made to Lewison et al. (2002)’s original critical literacy objectives to adapt them to the context of AI.
* Indicates learning objectives that are generated through synthesis.

in the our literature sample. We present a combination of preexisting and newly agreed upon learning objectives for each dimension of Lewison et al.’s (2002) framework. A detailed overview of where in the learning activities we identified these learning objectives can be found in B. One can use this overview as more concrete inspiration for carrying out the critical AI literacy learning objectives.

We organized the learning objectives according to the revised Bloom’s taxonomy (Anderson & Krathwohl, 2001). Its six levels each demand progressively higher levels of complexity and ordered thinking. Our analysis focuses on the final three levels. *Analyze*, level four, refers to separating material into its constituent parts and identifying their interrelations with one another and the overall structure. *Evaluate*, level five, involves students making judgments grounded in criteria. The uppermost level, *create*, requires students to combine elements to generate a novel, coherent whole.

4.5.1. *Disrupting the commonplace* (Table 6)

We conceptualized this dimension of critical AI literacy as giving children the ability to critique current and prospective applications of AI, societal perceptions of AI, and providing opportunities for critical identity work. It encourages learners to adopt a consumer perspective on AI (Monroy-Hernández & Resnick, 2008). This requires a student to recognize personal assumptions regarding AI, understand AI as a historical product capable of perpetuating past issues, and how tools mediate behavior (Verbeek, 2006).

Analyze. The practice in this dimension might start with analysis of personal AI-related usage patterns. To problematize these patterns, children have to gain an understanding of how these technologies are applied in their daily life. However, children often do not recognize applications of AI in their daily life. To prompt student inquiry, educators might ask students directly to identify tools that incorporate AI technology (Schaper et al., 2023; Solyst, Xie et al., 2023; Solyst, Yang et al., 2023). Additionally, questioning may be indirectly encouraged by leveraging the learners’ experiential knowledge, for instance, by incorporating technologies commonly used by children into learning activities (DiPaola et al., 2020; Lee et al., 2021, 2022; Rankin & Henderson, 2021).

A fundamental aspect of cultivating CL involves problematizing established knowledge (Shor, 1987). Consequently, children must understand their views are not solely their own but may also be influenced by peers, popular media and society (Fosnot, 1996). In creating a specific learning objective dedicated to interrogating personal conceptions about AI, we aimed to combine learner-centered analysis with social analysis, an integral component of critical literacy (Luke, 2013). It highlights the necessity of questioning our everyday rhetoric that has shaped the way we think about AI. To support learners in interrogating

Table 7
Critical AI Literacy learning objectives for the ‘considering multiple viewpoints’ dimension.

Considering multiple viewpoints	
Taxonomy	Learning objective
<i>Analyze</i>	Analyze how groups are impacted differently by AI technologies* Pay attention to or seek out voices of those who have been silenced or marginalized regarding AI Make difference visible
<i>Evaluate</i>	Reflect on multiple, contradictory and/or competing perspectives regarding AI Engage with/look through the lens of diverse cultural resources regarding AI*
<i>Create</i>	Create counternarratives to dominant discourses

Annotation. *Italicized* words indicate changes that were made to Lewison et al. (2002)’s original critical literacy objectives to adapt them to the context of AI.
* Indicates learning objectives that are generated through synthesis.

personal and societal assumptions of AI, educators can unveil new and previously hidden information about AI. This may include exposing hidden features or unique attributes of AI systems and their critical implications (Dai et al., 2023; Krakowski et al., 2022; Lee et al., 2022; Ng et al., 2022; Salac et al., 2023; Schaper et al., 2023).

Encouraging children to analyze how their interactions with AI shape behavior and identity is important for supporting meaningful critical identity work (Masterman, 1985). Learners can explore the relationship between personal, affective responses to AI and their relationship to self-identity and broader societal and ideological concerns (Mott et al., 2022; Schaper et al., 2022). They can envision how AI technologies can be altered to reflect their personal identity (Arada et al., 2023; Buddemeyer et al., 2022; Rankin & Henderson, 2021) or speculate on how AI might affect their future identities (Harrington & Dillahunt, 2021) or places in society as workers (Lee et al., 2021). As with any tool that enables people to accomplish new tasks, AI will influence behavior. This can be analyzed on a personal level (Ali et al., 2021; Lee et al., 2021, 2022; Williams et al., 2023) as well as from a social perspective (Ali et al., 2021; Krakowski et al., 2022; Williams et al., 2023). After exploring AI’s role in daily life, questioning current AI knowledge, and recognizing its influence on identity and behavior, learners can critically assess AI’s effect on societal progress and norms (Lee et al., 2021). As aforementioned, AI is both a product of history and the vehicle that shapes it. This reciprocal relationship underscores the significant impact AI may have in the future.

Evaluate. CL explicitly aims to critique dominant ideologies, cultures, economies, institutions, and political systems (Luke, 2012). Once children have analyzed how AI is applied in different systems and its potential impacts on identity, behavior and society, they are better equipped to discuss and critique current (Arada et al., 2023; Arastoopour Irgens et al., 2022; Everson et al., 2022; Forsyth et al., 2021; Lee et al., 2021, 2022; Solyst et al., 2022; Solyst, Xie et al., 2023; Solyst, Yang et al., 2023) and AI’s hypothetical applications (Bilstrup et al., 2020; Charisi et al., 2021; Kaspersen et al., 2022; Lin et al., 2023; Skinner et al., 2020), including designs created by peers (Schaper et al., 2022).

Create. This critique could provoke reflection on personal AI-related practices and guide learners to reassess, reflect, and renew their own engagement with AI. Thus, we propose a new learning objective that requires learners to formalize their reflections through the definition of personal norms and/or responsibilities towards AI use (Schaper et al., 2022).

4.5.2. *Considering multiple viewpoints* (Table 7)

We conceptualized this dimension as giving learners the ability to consider alternative perspectives, cultures, values and ways of being, as well as the tensions arising from these differences. To do this, learners

Table 8
Critical AI Literacy learning objectives for the ‘focusing on the sociopolitical’ dimension.

Focusing on the sociopolitical	
Taxonomy	Learning objective
Analyze	Study the impact that AI makes on the larger sociocultural landscape* Study the relationship between AI and power Question which people have been missing in the development of AI*
Evaluate	Critique ways in which AI creates or upholds unequal power relationships
Create	Challenge the power relationships that are created or upheld by AI Increase opportunities for subordinate groups to participate in society and AI

Annotation. *Italicized* words indicate changes that were made to Lewison et al. (2002)’s original critical literacy objectives to adapt them to the context of AI.
* Indicates learning objectives that are generated through synthesis.

must distinguish between different societal groups, consider invisible pressures they face, and empathize with their perspectives. This dimension is committed to inclusion, ensuring representation of the working class, cultural minorities, and other marginalized identities.

Analyze. In order to comprehend alternative perspectives, learners have to gain a nuanced understanding of AI’s impacts. Students can analyze how the same technology can have varying affects for different demographic groups (Arada et al., 2023; Buddemeyer et al., 2022; Solyst et al., 2022; Williams et al., 2023) through real-world case analysis, simulations or media. In their analysis, learners can pay attention to, consider, or seek out the voices that have been marginalized regarding AI development (Arada et al., 2023; Arastoopour Irgens et al., 2022; Everson et al., 2022; Lee et al., 2022; Solyst et al., 2022; Williams et al., 2023) or leverage their personal experience as marginalized group (Buddemeyer et al., 2022; Harrington & Dillahun, 2021; Rankin & Henderson, 2021). Visualization techniques may help learners unpack and understand the perspectives of different groups by contrasting personal opinions to that of others (Buddemeyer et al., 2022), comparing or contrasting the values and needs of various stakeholders (DiPaola et al., 2020; Lee et al., 2021; Schaper et al., 2022; Williams et al., 2023), or by embodying these different perspective through role playing different characters to highlight views, interests, and personal risks regarding AI technologies (Aki Tamashiro et al., 2021; Lin et al., 2023).

Evaluate. Through simultaneously engaging with multiple perspectives, learners can begin to unpack contradictions, competing viewpoints and the tensions between those narratives. AI has different stakeholders involved in its implementation. Learners can evaluate both existing and conceptual AI applications through multi-stakeholder analysis (Bilstrup et al., 2020; Lin et al., 2023; Schaper et al., 2022) and visualize tensions between stakeholder values by contrasting them (DiPaola et al., 2020; Lee et al., 2021; Schaper et al., 2022; Williams et al., 2023). Cultural values and perspectives influence the creation and use of AI. Learners can examination how AI is perceived and applied in different cultures through culturally significant case studies (Schaper et al., 2022), or embed the ideation process of new AI technologies within culturally significant design briefs (Schaper et al., 2020). However, for evaluations to be critical, learners must not only consider values, wants and needs, but also possible harmful consequences.

Create. Through counter-hegemonic critique, learners can create counternarratives to challenge dominant discourses. They can envision new possible worlds that are different from our current future trajectory based on these multiple viewpoints (Arada et al., 2023; Everson et al., 2022; Harrington & Dillahun, 2021; Schaper et al., 2023).

Table 9
Critical AI Literacy learning objectives for the ‘focus on the sociopolitical’ dimension.

Taking action	
Taxonomy	Learning objective
Analyze	Question AI creators’ processes and intentions*
Evaluate	Evaluate ethical implications of self-ideated conceptual AI systems* Consider accountability and define ethical responsibilities in relation to decisions made by, for, or with the assistance of AI*
Create	Appropriate or design AI to enhance everyday life Ideate, design, and/or create AI that take ethical considerations into account* Engaging in Praxis-reflection and action on the world in order to transform it

Annotation. *Italicized* words indicate changes that were made to Lewison et al. (2002)’s original critical literacy objectives to adapt them to the context of AI.
* Indicates learning objectives that are generated through synthesis.

4.5.3. Focusing on the sociopolitical (Table 8)

We conceptualized this dimensions as critique of domination, transformation of codes of power, and the creation of bridges to access AI technologies and their benefits. Learners must go beyond the individual and understand the broader sociopolitical context, including social norms, power dynamics, and barriers to access AI technologies, to participate in this critique.

Analyze. By considering and studying how AI impacts society at large (Forsyth et al., 2021; Kaspersen et al., 2022; Lin et al., 2020, 2023), learners can gain fundamental understandings that enable them to engage in the analysis, critique, and Praxis within this dimension. Through multimedia sources, they can observe and discuss how AI creates and reinforces power dynamics (Arada et al., 2023; Arastoopour Irgens et al., 2022; DiPaola et al., 2020; Everson et al., 2022; Forsyth et al., 2021; Harrington & Dillahun, 2021; Lee et al., 2021, 2022; Salac et al., 2023; Schaper et al., 2022; Solyst et al., 2022; Solyst, Xie et al., 2023; Williams et al., 2023) and how it might be used by those in power to strengthen their authority (Aki Tamashiro et al., 2021; Ali et al., 2021; Bilstrup et al., 2020; Kaspersen et al., 2022; Krakowski et al., 2022; Lee et al., 2021, 2022; Mott et al., 2022; Schaper et al., 2022; Solyst, Yang et al., 2023). As such, they can start dissecting power dynamics by considering who is missing from the design and development of AI (Solyst, Xie et al., 2023), questioning why certain voices are excluded from the process and how this reinforces inequities.

Evaluate. Learners may challenge the legitimacy of these unequal power relationships and make judgements based in sociopolitical considerations which can be formalized by choices made in simulations (Bilstrup et al., 2020; Kaspersen et al., 2022; Krakowski et al., 2022) or by guided discussion and debate positions (DiPaola et al., 2020; Salac et al., 2023).

Create. Learners can apply newfound perspectives to assert a stance regarding AI’s role in power imbalances by devising strategies to mitigate its impact (Solyst, Yang et al., 2023) or by producing tools to educate others about the perpetuated harms and power dynamics (Lee et al., 2022).

Additionally, learners can create ways to foster societal liberation and nurture opportunities for marginalized groups to gain access to otherwise inaccessible tools by democratizing AI technology, for instance, through the development of ways to educate others on how to use AI (Lee et al., 2022).

4.5.4. Taking action (Table 9)

We conceptualized this dimension as transitioning from critical analysis to social action – Praxis – with an explicit emphasis on promoting social justice regarding AI. Unlike previous dimensions centered on deconstructing AI, this dimension involves learners taking on a

producer role to (re)design socially, culturally, politically, and ethically aware AI applications. Nonetheless, Bilstrup et al. (2020) highlight that when engaging youth in AI design concepts, design alone is insufficient to foster critical AI dispositions.

Analyze. In traditional literacy, readers analyze authorial intent by unpacking tone, point of view, and intended audience. For critical AI literacy, learners could analyze the intent of AI creators by dissecting their processes (Arada et al., 2023; Arastoopour Irgens et al., 2022; Everson et al., 2022; Solyst et al., 2022; Williams, Machado, Druga, Breazeal, & Maes, 2018), design choices (Lee et al., 2022; Solyst, Yang et al., 2023), or by tracing their steps through redesign activities (DiPaola et al., 2020; Lee et al., 2021; Williams et al., 2023).

Evaluate. Moreover, it is not only necessary for learners to critically evaluate the choices made by those who have created existing AI technologies, but they should also be able to evaluate the consequences of their own conceptual AI systems (Bilstrup et al., 2020; DiPaola et al., 2020; Solyst et al., 2022; Solyst, Xie et al., 2023). Learners may also materialize critical consciousness by considering who is accountable for decisions made *by*, *for*, or *with* AI tools. For instance, children can discuss or define AI policy (Ali et al., 2021), define who is responsible for fixing an unfair AI system and/or their outcomes (DiPaola et al., 2020; Solyst, Xie et al., 2023) or define rules for fairer AI technologies (Rankin & Henderson, 2021; Salac et al., 2023).

Create. AI technologies offer affordances that can be used to address societal challenges. Through user-centered design practices, learners can identify, adapt or design AI technologies to address these challenges (Arastoopour Irgens et al., 2022; Ng et al., 2022). Learners can actualize outcomes from prior critical inquiry by defining design requirements and creating AI concepts, redesigns of AI systems, or ML models that incorporate critical considerations (Bilstrup et al., 2020; DiPaola et al., 2020; Krakowski et al., 2022; Lee et al., 2021; Williams et al., 2023).

Lastly, Praxis goes beyond the learning context and aims to actualize change and transformation in the world. The literature suggests teaching learners skills they can use in their daily life to challenge AI system harms, such as identifying and addressing deepfake images and videos (Ali et al., 2021; Williams et al., 2023) or ways to trick facial recognition (Aki Tamashiro et al., 2021; Lee et al., 2022).

5. Discussion

5.1. Towards a framework for critical AI literacy

Over the past three years, there has been a rise in studies exploring activities to promote youths' critical reflections on AI's societal, political, cultural, and ethical implications. Through the lens of CL, we investigated how these studies sought to support learner's capacity to analyze, critique, and transform AI's critical implications. Starting with Lewison et al. (2002)'s framework, we synthesized 13 additional learning objectives with respect to critical AI literacy. The newly formulated learning objectives focus primarily on deconstructing AI's use and development. Our proposed framework integrates concepts from critical computational literacy (Lee & Soep, 2016) and critical digital literacy (Pangrazio, 2016). Both conceptual frameworks recognize dichotomous roles of consumer/user and producer within multimodal and transmedia artifacts, which seem to be increasingly confluent. The AI context necessitates an understanding of both the influences constructing the artifacts and our interpretations as users. Therefore, our model presents learning objectives reflecting on AI's critical implications on cultural, societal, and political structures, self-impact, and analyzing AI artifact development, whether created by the learner or others. Our intention is not to supersede existing CL works in the realm of multimodal and transmedia artifacts, but to build upon them. In doing so, we aimed to provide practical support for educators

and researchers interested in creating critical pedagogy by synthesizing learning objectives for different levels of complexity. Like how Lewison et al. (2002) asked "what is critical literacy?" and "what does it look like in the classroom?", we set the first steps toward answering: "what is critical AI literacy, and what can it look like in education?". We suggest continued conceptualization of critical AI literacy.

Similar to how Ng et al. (2021) categorized AI literacy to achieve a more comprehensive conceptualization, we also adopted Bloom's cognitive levels to map how existing literature has attempted to foster critical AI literacy. Though not common in critical pedagogy, we believe this intertwining of CL with a cognitive taxonomy to be practical, since it acknowledges learner background knowledge and capacities. Developing criticality constitutes prerequisite knowledge, but also cognitive capacities (Corson, 1984). Prior work indicates that reflective judgment abilities form through a developmental sequence (King & Kitchener, 2004). As such, the hierarchical levels of Bloom's taxonomy can support the sequencing of learning activities according to learners' abilities. Consistent with Computational Empowerment (Iversen et al., 2018), we advocate for criticality's integration into broader AI pedagogy, encompassing both technical understanding and responsible AI usage.

We emphasize that learners do not need technical skills to engage in critical critique of AI. However, maturing from basic recognition of AI technology to critiquing critical implications requires a cognitive understanding that builds on the lower-order thinking skills from Bloom's taxonomy (Wong, Ma, Dillenbourg, & Huan, 2020). Consequently, we need to conduct further research to determine what knowledge, dispositions, and developmental aspects are essential for practicing critical AI literacy across all six levels of Bloom's taxonomy, creating a broader developmental template often not found in critical pedagogies (Luke, 2013).

Prior research delineates CL into two educational perspectives: as a skill and as a mode of acting (Luke, 2013). While our approach aligns with Bloom's cognitive model, we maintain that critical AI literacy embodies a disposition and a mode of doing. Our literature corpus includes diverse study designs. While our aim is to develop a research-based critical AI pedagogy, we do not believe quantitative evaluation is mandatory for studying critical AI literacy. In our view, researchers should adapt analytical tools used to study CL to examine the effectiveness of newly developed critical AI literacy learning activities (Rogers & O'Daniels, 2015).

5.2. Future opportunities for critical AI literacies in CCI

When Lewison et al. (2002) described their model, they stressed the interrelatedness of the four dimensions. Through our framework synthesis, we identified how the learning objectives were represented in the activities of our literature corpus. In few studies, learners were involved in activities that supported learning objectives across all dimensions. However, design without critical inquiry can yield ethically questionable outcomes (Bilstrup et al., 2020), and neglecting sociopolitical analysis when considering multiple viewpoints can lead to unfounded ethical judgments. By presenting this framework, we aim to encourage the sequencing of learning activities across various dimensions and complexity levels, while also inspiring the design of new critical AI literacy activities. We will highlight potential perspectives and methods that can be used as a springboard for future learning activity design.

Broadening perspectives. As per Lewison et al. (2002)'s original learning objectives, we ask learners to question the Zeitgeist surrounding AI and how the technology disrupts society. Learning objectives in 'disrupting the commonplace' are interconnected and should encourage analysis on personal, societal, and historical levels. The 'commonplace' evolves alongside the emergence of new technologies, introducing new critical implications and societal narratives. Hence, it is vital to consider AI's diverse impacts. For instance, it is possible that the rapid adoption of

new generative AI-based systems might transform our current attention economy into an economy of intimacy through personal AI (Edwards & Edwards, 2024). As such, a greater need for children to be emotionally aware when interacting with these AI technologies is being created.

While most work was conducted in North America, we recognize that values vary across cultures and countries, especially those with extensive AI integration and mandatory curricula (Dai et al., 2023), reflecting culturally disparate perspectives on the future of AI. As seen in the work of Schaper et al. (2020, 2022), different perspectives offer rich opportunities for cultural reflection. Cultural adaptation, like using culturally relevant stories (Charisi et al., 2021), can also facilitate understanding of abstract concepts and resonate with children from multifaceted backgrounds. Additionally, to develop critical consciousness, young learners need guidance to challenge culturally dominant narratives, including Western and colonial approaches (Arada et al., 2023).

Accessibility entails recognizing and eliminating barriers hindering full inclusive participation. While some publications highlighted the necessity of having diverse entry points (Arada et al., 2023; Schaper et al., 2023), it remains unclear how this applies to individuals with varying abilities and socioeconomic statuses. Despite disproportionate impacts from AI (Buolamwini & Gebru, 2018), marginalized groups are often neglected in STEM discussions (Cech, 2022). Critical AI literacy holds significant potential here, as it may offer tools for marginalized learners to comprehend and resist AI-driven oppression. While our review showcased several examples of activities informed by Black-feminist thought (Collins, 2008); this could be extended by leveraging a range of philosophical lenses to guide the design of learning activities, such as decolonial approaches (Zembylas, 2023), queer, crip, and feminist theories (Bardzell, 2010; Light, 2011; McRuer, 2008), post-humanist philosophy (Barad, 2003; Haraway, 2006), and community-led practices (Costanza-Chock, 2020).

The critical implications of AI identified in our review primarily focused on human-centered perspectives. This overlooks the viewpoints of more-than-human perspectives (Whatmore, 2006). AI will impact not only humans, but also their relationships with the world, animals, and non-human objects. For instance, using AI, robotics, or virtual reality animals as entertainment may perpetuate power imbalances between animals and humans. Discussions of the environmental costs associated with the training and running of AI models is a good place to begin from an ecological standpoint (Lee et al., 2021). There are, however, other ecological impacts that can be examined, including the whole chain of obtaining raw materials to the disposing of technology required to build AI systems, as well as who regulates these infrastructures and who is affected by them (Crawford & Joler, 2018; Gourlet, Ricci, & Crépel, 2024). Future learning activities can take inspiration from Leander and Burris (2020)'s post-human CL to address these perspectives.

Creating new pathways. Dialogic learning is one of the hallmarks of CL education (Luke, 2013). Accordingly, we identified some structured dialogue-based activities in our review (Dai et al., 2023; DiPaola et al., 2020; Lin et al., 2023; Salac et al., 2023; Skinner et al., 2020). In particular, Lin et al. (2023) scaffolds students' moral reasoning regarding autonomous vehicles via a guided debate technique. Debate is crucial for exposing oneself to different viewpoints and making space for radical ideas (Lo, 2017). Sociopolitically-centered dialogic learning activities need structures that allow conflict to surface (Frauenberger, Spiel, Scheepmaker, & Posch, 2019). As such, more opportunities arise to deepen students' critical engagement with AI through agnostic approaches to dialogue (Suransky & Alma, 2018).

The discourse surrounding AI sociotechnical perspectives often fluctuates between the potential benefits and the potential harms. We identified several learning activities that asked learners to identify harms and benefits of AI technologies (Harrington & Dillahunt, 2021; Schaper et al., 2020, 2023; Skinner et al., 2020; Solyst et al., 2022; Williams et al., 2023). While the utopia versus dystopia and harmful

versus beneficial dichotomies may initiate reflection, it is essential to emphasize the nuances, complexities, and trade-offs in the social and ethical impacts of AI (Foucault, 1970). We propose that future activities embrace the entanglements of our sociotechnical world (Frauenberger, 2019). It is also important to note, an over-reliance on deterministic thinking may emphasize the idea that impacts of AI are eventualities and may negatively affect participants' perceived ability to see possibilities for change (Aleman & Martinez, 2024).

Solely focusing on dystopian narratives of AI may discourage and undermine learners (Sudmann, 2019). Introducing playful elements into critical AI literacy learning activities may assist children's motivation to engage in critique, as well as promote divergent thinking (Proyer, Tandler, & Brauer, 2019). For instance, Malpass (2013) proposes a cut-up and reassemble method for critical design that could allow students to critique existing AI products through exaggeration, absurdity, and parody. While it is crucial to help children foster critical dispositions, it is equally important to provide them with tools to act on those dispositions (Stamboliev, 2023). Engaging children in design, Praxis and acts of transformation fosters a sense of agency and change. We identified various design paradigms that informed constructionist learning activities. Iivari et al. (2022) highlight further critical design approaches which can shape future critical AI literacy learning activities. Herein we stress the importance of exploring ways to facilitate critical reflection during different stages of the design and making process, for instance by combining dialogic, experiential, and hands-on approaches (Antle et al., 2022).

5.3. Infrastructure of critical AI literacy learning

Following Stamboliev (2023), we believe critical pedagogy on AI should not be confined to the computer science curriculum. For instance, Kaspersen et al. (2022) implemented a module on ML for predicting democratic election's voting outcomes in a civics course. Critical AI literacy should be understood as a cross-curricular project.

Smith et al. (2023) advocate for participatory practices to embed Computational Empowerment into K-12 education and beyond, a stance we share. In our review, we observed limited support for co-facilitation, despite a growing body of work co-creating AI curricula with teachers (Gardner-McCune et al., 2023; Lin & Van Brummelen, 2021; Sperling, Stenliden, Nissen, & Heintz, 2024; Yau et al., 2022). For realizable curriculum integration, future efforts should aid educators in developing their own critical stances and pedagogical strategies for teaching critical AI literacy.

Lee and Soep (2016) recognize the challenges of authentic learning as intention may be questioned by excessive projection of views by educators, leading to students mirroring their opinions. In contrast to rationalism, critical pedagogy does not presume to find truth through critique (Luke, 2013). The aim is to give learners the tools to form their own perspectives, rather than imposing a single viewpoint. It is therefore crucial for learners to have agency in their learning trajectory through regulation of learning, as demonstrated in various studies (Aki Tamashiro et al., 2021; Arastoopour Irgens et al., 2022; Everson et al., 2022). Providing the opportunity for learners to regulate and evaluate their own learning facilitates an environment where students can enact agency (Bolton & Elmore, 2013; Hadwin, Järvelä, & Miller, 2017).

6. Conclusion

At the time of writing, new generative AI systems are released almost weekly. Considering their disruptive potential, there is an urgent need to support young people's critical perspectives toward these systems. Thus, we conducted a scoping review and framework synthesis (Carroll et al., 2013) targeting records detailing activities supporting young people's reflection on AI's social, political, cultural, and ethical impacts. In our review, we found a supportive foundation for

developing a research-based critical pedagogy for AI. We termed this *critical AI literacy*: the capacity to analyze, critique, and transform AI’s critical implications. To operationalize critical AI literacy for broader educational implementation, we outlined its integration across common curriculum elements: content, learning activities, and learning objectives based in critical literacy theory (Lewison et al., 2002). In addition to this literature-based overview, we identify additional directions for research, design, and education. Finally, we want to give acknowledgement to other inspiring works that showcase the broader critical turn of the CCI community (Antle et al., 2022; Cumbo & Iversen, 2020; Dindler, Smith, & Iversen, 2020; Gazulla et al., 2024; Iivari et al., 2024, 2022; Kinnula & Iivari, 2021; Morales-Navarro & Kafai, 2023; Sharma et al., 2024; Smith et al., 2023; Spiel, Frauenberger, Keyes, & Fitzpatrick, 2019; Van Mechelen, Have Musaeus, Iversen, Dindler, & Hjorth, 2021; Leena Ventä-Olkkonen et al., 2024; Ymous et al., 2020). We hope our work can inspire future investigations on critical pedagogy for technology education.

7. Selection and participation

No children were involved in the development of this work.

CRediT authorship contribution statement

Annemiek Veldhuis: Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis, Data curation, Conceptualization, Validation. **Priscilla Y. Lo:** Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation, Validation, Conceptualization. **Sadhbh Kenny:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization, Validation. **Alissa N. Antle:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Table A.10
Coding framework derived from Lewison et al. (2002)’s critical literacy framework.

Disrupting the commonplace		
Taxonomy	Learning objective	Description
Analyze	Question “everyday” ways of seeing AI	Activities in which learners analyze how the general public sees AI technologies
	Analyze how AI technologies have changed society’s trajectory and the status quo	Activities in which learners reflect on how AI has impacted society compared to the past
	Study AI to analyze how it shapes identity	Activities in which learners reflect on their own feelings regarding AI or how AI might shape their own behavior and life
	Question how people’s behaviors are constructed by AI technologies	Activities in which learners analyze how current AI technologies might drive people to behave differently
Considering multiple viewpoints		
Analyze	Pay attention to or seek out voices of those who have been silenced or marginalized regarding AI	Activities in which learners hear the experiences and opinions of marginalized groups regarding AI or activities in which learners voice their own experiences regarding being marginalized through/by AI (ageism, racism, gender discrimination and other intersectionalities)
	Make difference visible	Activities in which children visualize the different perspectives/needs/requirements regarding AI
Evaluate	Reflect on multiple, contradictory and/or competing perspectives regarding AI	Activities in which learners reflect on value tensions and competing narratives regarding AI
Create	Create counternarratives to dominant discourses	Activities in which learners challenge dominant discourses and create narratives around new possible worlds.
Focusing on the sociopolitical		
Analyze	Study the relationships between AI and power	Activities in which learners learn about or experience power relationships/authority, algorithmic bias and the perpetuation of stereotypes caused by, perpetuated by or enforced by AI
Evaluate	Critique ways in which AI creates or upholds unequal power relationships	Activities in which learners judge power relationships/authority, algorithmic bias and the perpetuation of stereotypes caused by, perpetuated by or enforced by AI

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Alissa N. Antle reports financial support was provided by Government of Canada Social Sciences and Humanities Research Council. Annemiek Veldhuis reports financial support was provided by Government of Canada Social Sciences and Humanities Research Council. Sadhbh Kenny reports financial support was provided by Government of Canada Social Sciences and Humanities Research Council. Priscilla Y. Lo reports financial support was provided by Government of Canada Social Sciences and Humanities Research Council. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Coding framework

See Table A.10.

Appendix B. Critical AI literacy framework extended

See Tables B.11–B.14.

(continued on next page)

Table A.10 (continued).

<i>Create</i>	Challenge the power relationships created/upheld by AI	Activities about resisting or educating others about certain AI technologies that reinforce power dynamics
	Increase opportunities for subordinate groups to participate in society and AI	Activities in which students envision ways for others to access AI technologies
Taking action		
<i>Create</i>	Engaging in Praxis-reflection and action on the world in order to transform it	Activities in which students learn critical skills that they can use beyond the educational environment or engage in action beyond the educational environment

Table B.11

Critical AI Literacy learning objectives for the ‘disrupting the commonplace’ dimension.

Disrupting the commonplace		
Taxonomy	Learning objective	Activity quality
<i>Analyze</i>	Question how AI is integrated in their daily life*	- Identify products that incorporate AI technology (Schaper et al., 2023; Solyst, Xie et al., 2023; Solyst, Yang et al., 2023) - Focus on AI tech that children use on a regular basis (DiPaola et al., 2020; Lee et al., 2021, 2022; Rankin & Henderson, 2021)
	Question preconceptions of AI and its critical implications*	- Reveal hidden aspects of AI its critical implications (Krakowski et al., 2022; Lee et al., 2022; Ng et al., 2022; Salac et al., 2023) - Highlight unique attributes of AI and their critical implications (Dai et al., 2023; Schaper et al., 2023)
	Question “everyday” ways of seeing AI	Provide new perspectives on AI and its impact on the world (Arada et al., 2023; Arastoopour Irgens et al., 2022; Everson et al., 2022; Forsyth et al., 2021; Lee et al., 2021, 2022; Schaper et al., 2022; Solyst et al., 2022; Solyst, Xie et al., 2023; Solyst, Yang et al., 2023; Williams et al., 2023)
	Study AI to analyze how it shapes identity	- Identify personal feelings regarding AI technologies (Mott et al., 2022; Schaper et al., 2022) - Focus on how AI technology can be altered to reflect personal identity (Arada et al., 2023; Buddemeyer et al., 2022; Rankin & Henderson, 2021) - Identify how AI might influence their future place in society (Harrington & Dillahun, 2021; Lee et al., 2021)
	Question how people's behaviors are constructed by AI technologies	- Analyze how interactions with AI might influence personal behavior (Ali et al., 2021; Lee et al., 2021, 2022; Williams et al., 2023) - Analyze the opportunities, capabilities, and risks that AI technologies present to individuals (Ali et al., 2021; Krakowski et al., 2022; Williams et al., 2023)
	Analyze how AI technologies disrupt society's trajectory and status quo	Analyze cases of past applications of AI and whether they have made societal changes (Lee et al., 2021; Lin et al., 2023)
<i>Evaluate</i>	Judge how AI is applied in daily life*	- Discuss issues of existing AI technologies (Arada et al., 2023; Arastoopour Irgens et al., 2022; Everson et al., 2022; Forsyth et al., 2021; Lee et al., 2021, 2022; Solyst et al., 2022; Solyst, Xie et al., 2023; Solyst, Yang et al., 2023) - Ask students to minimize bias in ML models (Krakowski et al., 2022)
	Judge potential applications of AI*	- Discuss implications of potential, future AI technologies (Bilstrup et al., 2020; Charisi et al., 2021; Kaspersen et al., 2022; Lin et al., 2023; Skinner et al., 2020) - Critique designs created by peers (Schaper et al., 2022)
<i>Create</i>	Envision responsible use of AI technologies*	Define personal responsibilities in making choices around AI use (Schaper et al., 2022)

Annotation. *Italicized* words indicate changes that were made to Lewison et al. (2002)'s original critical literacy objectives to adapt them to the context of AI.

* Indicates learning objectives that are generated through synthesis.

Table B.12

Critical AI Literacy learning objectives for the ‘considering multiple viewpoints’ dimension.

Considering multiple viewpoints		
Taxonomy	Learning objective	Activity quality
<i>Analyze</i>	Analyze how AI is impacting different aspects of people's daily life*	- Incorporate analysis of real-world cases (Ali et al., 2021; Arada et al., 2023; Everson et al., 2022; Krakowski et al., 2022; Lee et al., 2021, 2022; Salac et al., 2023; Schaper et al., 2022; Solyst, Xie et al., 2023; Solyst, Yang et al., 2023; Williams et al., 2023) - Use simulations to show the greater impact of AI (Ali et al., 2021; Krakowski et al., 2022; Williams et al., 2023) - Analyze media that highlight ethical implications (Arada et al., 2023; Arastoopour Irgens et al., 2022; Everson et al., 2022; Solyst et al., 2022; Williams et al., 2023) - Take a multistakeholder perspective (DiPaola et al., 2020; Lee et al., 2021; Schaper et al., 2022; Williams et al., 2023)

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Table B.12 (continued).

Considering multiple viewpoints		
Taxonomy	Learning objective	Activity quality
<i>Evaluate</i>	Analyze how groups are impacted differently by AI technologies*	- Analyze one AI technology and its effect on different groups (Arada et al., 2023; Buddemeyer et al., 2022; Solyst et al., 2022; Williams et al., 2023) - Incorporate analysis of real-world cases (Ali et al., 2021; Arada et al., 2023; Everson et al., 2022; Krakowski et al., 2022; Lee et al., 2021, 2022; Salac et al., 2023; Schaper et al., 2022; Solyst, Xie et al., 2023; Solyst, Yang et al., 2023; Williams et al., 2023)
	Pay attention to or seek out voices of those who have been silenced or marginalized <i>regarding AI</i>	- Read about or listen to the experiences of marginalized groups regarding AI (Arada et al., 2023; Arastoopour Irgens et al., 2022; Everson et al., 2022; Lee et al., 2022; Solyst et al., 2022; Williams et al., 2023) - Leverage personal experience as marginalized group (Buddemeyer et al., 2022; Harrington & Dillahun, 2021; Rankin & Henderson, 2021)
	Make difference visible	- Visualize different perspectives through contrasting personal opinion compared to others' (Buddemeyer et al., 2022) - Visualize values or needs of stakeholders (DiPaola et al., 2020; Lee et al., 2021; Schaper et al., 2022; Williams et al., 2023) - Roleplay different characters and focus on their views, interests and implications regarding AI tech (Aki Tamashiro et al., 2021)
	Reflect on multiple, contradictory and/or competing perspectives <i>regarding AI</i>	- Visualize tensions between different stakeholder values by contrasting them (DiPaola et al., 2020; Lee et al., 2021; Schaper et al., 2022; Williams et al., 2023) - Evaluate a design through a multistakeholder perspective (Bilstrup et al., 2020; Lin et al., 2023; Schaper et al., 2022)
	Engage with/look through the lens of diverse cultural resources regarding AI*	- Analyze how AI is seen and used in different cultures (Schaper et al., 2022) - Embed the ideation process of new AI technologies within culturally significant design briefs (Schaper et al., 2020)
<i>Create</i>	Create counternarratives to dominant discourses	Envision new possible worlds that are different from our current future trajectory based on various perspectives (Arada et al., 2023; Everson et al., 2022; Harrington & Dillahun, 2021; Schaper et al., 2023)

Annotation. *Italicized* words indicate changes that were made to Lewison et al. (2002)'s original critical literacy objectives to adapt them to the context of AI.

* Indicates learning objectives that are generated through synthesis.

Table B.13

Critical AI Literacy learning objectives for the 'focusing on the sociopolitical' dimension.

Focusing on the sociopolitical		
Taxonomy	Learning objective	Activity quality
<i>Analyze</i>	Study the impact that AI makes on the larger sociocultural landscape*	Discuss the potential impacts of AI on society at large (Forsyth et al., 2021; Kaspersen et al., 2022; Lin et al., 2020, 2023)
	Study the relationship between AI and power	- Engage with scenario's, media or speculative media that highlight the creation and perpetuation of power dynamics through AI (Arada et al., 2023; Arastoopour Irgens et al., 2022; DiPaola et al., 2020; Everson et al., 2022; Forsyth et al., 2021; Harrington & Dillahun, 2021; Lee et al., 2021, 2022; Mott et al., 2022; Salac et al., 2023; Schaper et al., 2022; Solyst et al., 2022; Solyst, Xie et al., 2023; Williams et al., 2023) - Investigate how AI might be used to strengthen authority (Aki Tamashiro et al., 2021; Ali et al., 2021; Bilstrup et al., 2020; Kaspersen et al., 2022; Krakowski et al., 2022; Lee et al., 2021, 2022; Schaper et al., 2022; Solyst, Yang et al., 2023)
<i>Evaluate</i>	Question which people have been missing in the development of AI*	Question who should be involved in the creation of future, fairer AI systems (Solyst, Xie et al., 2023)
	Critique <i>ways in which AI creates or upholds</i> unequal power relationships	- Situate simulations and ML model creation in sociopolitically relevant cases (Bilstrup et al., 2020; Kaspersen et al., 2022; Krakowski et al., 2022) - Formalize stances through discussion (Salac et al., 2023) or debate (DiPaola et al., 2020)
<i>Create</i>	Challenge the power relationships <i>that are created or upheld by AI</i>	- Generate ways to mitigate the creation and/or perpetuation of power relationships by AI (Solyst, Yang et al., 2023) - Produce tools that can educate others about the harms and power dynamics created by AI technologies (Lee et al., 2022)
	Increase opportunities for subordinate groups to participate in society <i>and AI</i>	Conceptualize or create educational tools (Lee et al., 2022)

Annotation. *Italicized* words indicate changes that were made to Lewison et al. (2002)'s original critical literacy objectives to adapt them to the context of AI.

* Indicates learning objectives that are generated through synthesis.

Table B.14
Critical AI Literacy learning objectives for the ‘taking action’ dimension.

Taking action		
Taxonomy	Learning objective	Activity quality
Analyze	Question AI creators’ processes and intentions*	• Show the AI creator’s process (Arada et al., 2023; Arastoopour Irgens et al., 2022; Everson et al., 2022; Solyst et al., 2022; Williams et al., 2023) • Analyze the design choices made by AI creators (Lee et al., 2022; Solyst, Yang et al., 2023) • Redesign an existing AI technology (DiPaola et al., 2020; Lee et al., 2021; Williams et al., 2023)
	Evaluate ethical implications of self-ideated conceptual AI systems*	• Use prompts to evaluate the ethical implications of self-ideated conceptual AI systems (Bilstrup et al., 2020; DiPaola et al., 2020; Solyst et al., 2022; Solyst, Xie et al., 2023)
Evaluate	Consider accountability and define ethical responsibilities in relation to decisions made by, for, or with the assistance of AI*	• Discuss or define potential AI policy (Ali et al., 2021) • Define who is accountable for fixing unfair AI systems and/or their outcomes (DiPaola et al., 2020; Solyst, Xie et al., 2023) • Define rules or design requirements for fairer AI technologies (Rankin & Henderson, 2021; Salac et al., 2023)
	Appropriate or design AI to enhance everyday life	• Identify societal challenges that could be tackled through AI technology (Arastoopour Irgens et al., 2022; Ng et al., 2022; Schaper et al., 2020; Solyst et al., 2022; Solyst, Xie et al., 2023)
Create	Ideate, design, and/or create AI that take ethical considerations into account*	• Use ethical analysis to define requirements for new AI concepts, redesign of AI systems or ML models (Bilstrup et al., 2020; DiPaola et al., 2020; Krakowski et al., 2022; Lee et al., 2021; Williams et al., 2023)
	Engage in Praxis-reflection and action on the world in order to transform it	• Learn how to engage in action beyond the study activities through applicable skills (Aki Tamashiro et al., 2021; Ali et al., 2021; Lee et al., 2022; Williams et al., 2023) • Create tools that can educate others (Lee et al., 2022)

Annotation. *Italicized* words indicate changes that were made to Lewison et al. (2002)’s original critical literacy objectives to adapt them to the context of AI.

* Indicates learning objectives that are generated through synthesis.

Data availability

Data will be made available on request.

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